



UNIT 03

PERIODIC TABLE & PERIODICITY OF PROPERTIES

Lithium



Pure lithium becomes dull when exposed to air.

Sodium



Pure sodium is a light silver color.

Potassium



Pure potassium is a light silver color.

Rubidium



Pure rubidium is a dark silver color.

Cesium



Pure cesium is a silver-gold color.

SHORT QUESTIONS

1. What is periodic table?**DEFINITION**

"A tabular chart of the chemical elements according to their atomic numbers so that elements with similar properties are in the same column."

2. Define periodicity?**DEFINITION**

"The reoccurrence of elements in the periodic table at regular interval is called periodicity of the elements."

3. Define triads?**DEFINITION**

"Dobereiner arranged similar elements in groups of three elements called triad or Dobereiner's Triads."

4. State and explain the law of triads?**DOBEREINER LAW OF TRIADS****INTRODUCTION:**

The German Chemist, Johann Dobereiner made the first significant effort in the development of the periodic table. It was first attempt towards classification of elements, it is also called the Dobereiner's law of Triads.

STATEMENT:

If elements are arranged in increasing order of their atomic masses, the atomic mass of the middle element of the triad is approximately the arithmetic mean of the other two.

ELEMENTS	ATOMIC MASS	AVERAGE
Lithium (Li)	6.9	$\frac{6.9 + 39.0}{2} = 22.95$
Sodium (Na)	23.0	
Potassium (K)	39.0	
Calcium (Ca)	40.1	$\frac{40.1 + 137.3}{2} = 88.65$
Strontium (Sr)	87.6	
Barium (Ba)	137.3	
Chlorine (Cl)	35.5	$\frac{35.5 + 126.9}{2} = 81.2$
Bromine (Br)	79.9	
Iodine (I)	126.9	

FAILURE:

All the known elements did not follow this rule. Law of triads was rejected as some triads nearly had same atomic masses, e.g., (Fe, Co, Ni), (Ru, Rh, Pd), (Os, Ir, Pt).

5. State and explain octave's law?

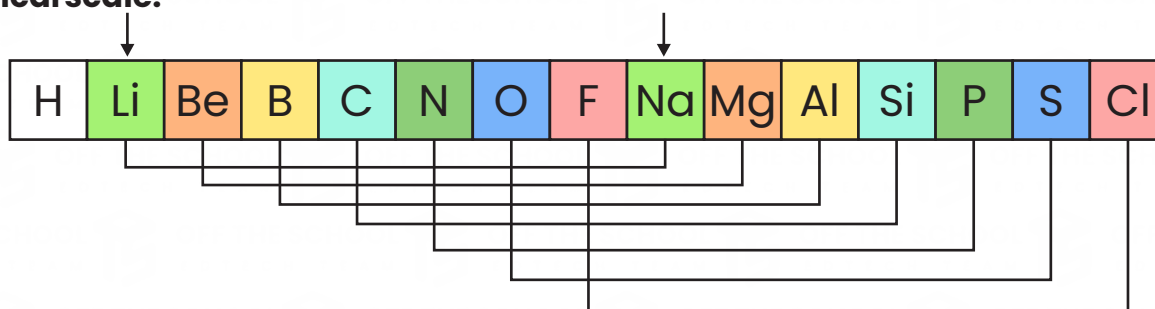
NEWLAND'S LAW OF OCTAVE

INTRODUCTION:

In 1864, the English Chemist, John Newland arranged the then known 56 elements in an increasing order of their atomic weights.

STATEMENT:

Newlands' law of octaves states that; when elements are arranged by increasing atomic mass, the properties of every eighth element repeat, just like the eighth note in a musical scale.



EXPLANATION:

When the lighter elements are arranged in order of their increasing atomic weights, then every eighth element is similar to the first element in its properties, similarly as the eighth note of a musical scale is similar to the 1st one. e.g. Na, 8th element resembles in their properties with Li.

Similarly, the 8th element with Na, and so on.

DEMERITS OR REASON TO FAILURE:

Newland's system of classification failed to explain similarities in higher elements.

6. What is the contribution of lother meyer in classification of periodic table.

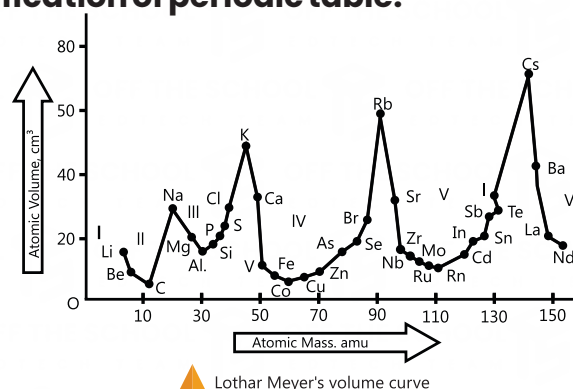
LOTHER MEYER'S CLASSIFICATIONS:

His classification includes 56 elements and their atomic volumes were plotted against their atomic weights obtaining similar peaks.

7. Determine the demarcation of periodic table into s, p, d and f block?

BLOCKS IN MODERN PERIODIC TABLE:

On the basis of the atomic structure and electronic configuration of the elements of periodic table concept of blocks is introduced. The periodic table is divided into four blocks s, p, d and f block.



8. What is Mendeleev's periodic law? Explain statement, merits, demerits of his periodictable?

MENDELEEV'S PERIODIC TABLE

INTRODUCTION:

Dmitri Ivanovich Mendeleev (1834–1907) gave a periodic law in 1869.

STATEMENT:

According to Mendeleev's periodic law: **"The properties of elements are a periodic function of their atomic masses."**

LONG QUESTIONS**1. Explain the modern periodic law?****MODERN PERIODIC TABLE AND LAW**

MOSELEY in 1913 formulated this law.

STATEMENT:

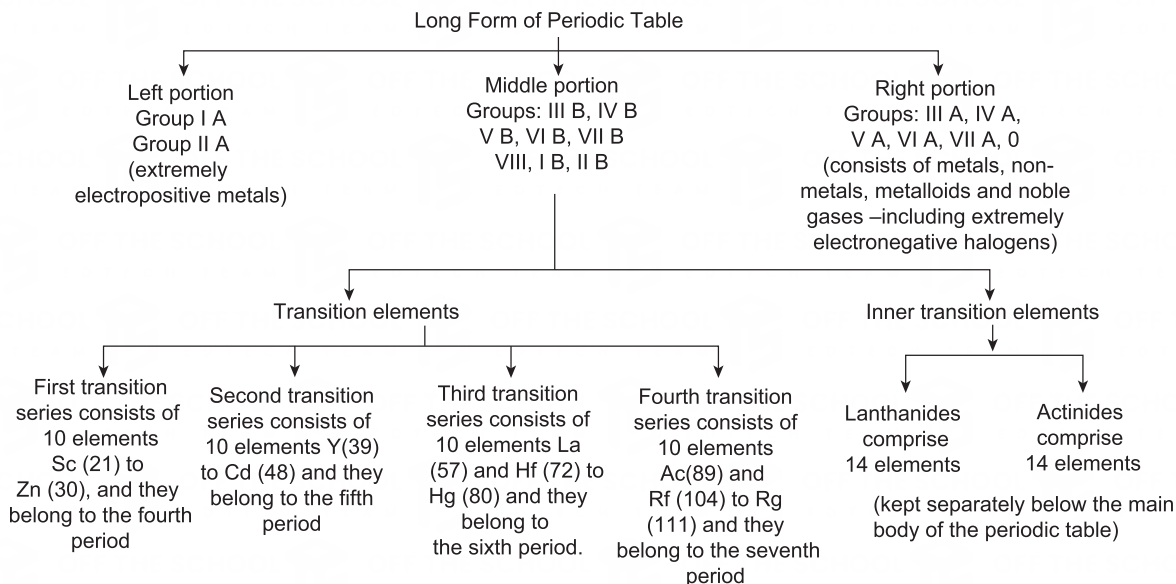
"The properties of the element are the periodic function of their atomic number."

- ▶ The standard for the selection of the group is valence electrons and that of period is the total number of shells.
- ▶ The criterion of the selection of the block (s,p,d,f) is the electronic configuration.
- ▶ Periods are the horizontal rows of periodic table.
- ▶ Groups are the vertical columns of periodic table.
- ▶ The first period contains only 2 elements (H and He) and it is shortest period.
- ▶ The second and third period contain 8 elements and it is 1st and 2nd short periods.
- ▶ The fourth and fifth period contain 18 elements and is called 1st and 2nd long periods.
- ▶ The sixth period contains 32 elements and it is known as the longest period.
- ▶ The seventh period is incomplete period of the periodic table.
- ▶ Group I-A is called Alkali metals OR Lithium family.
- ▶ Group II-A is called Alkaline earth metals OR Beryllium family.
- ▶ Group III-A is called Triels elements OR Boron family.
- ▶ Group IV-A is called Tetrels elements OR Carbon family.
- ▶ Group V-A is called Pnictogens elements OR Nitrogen family.
- ▶ Group VI-A elements are called Chalcogens elements OR Oxygen family.
- ▶ Group VII-A is called Halogens.
- ▶ Group VIII-A is called Noble gases or inert gases or zeroth group or aerogens.
- ▶ Elements of sub-group B or d-block elements are outer transition elements or metals.

LINK TO LEARNING

While the elements of f-block (lanthanides and actinides series) are inner transition elements or metals. Explore visualizations (<http://openstaxcollege.org/l/16pertrends>) of the periodic trends discussed in this section (and many more trends). With just a few clicks, you can create three-dimensional versions of the periodic table showing atomic size or graphs of ionization energies from all measured elements.

LINK 01



9. List and explain the types of elements based on electronic configuration?

TYPES OF ELEMENTS BASED ON ELECTRONIC CONFIGURATION

i. NOBLE GASES:

In periodic table zero group (VIII-A) elements are known as **noble gases**. They are colorless, inert (unreactive) and diamagnetic gases.

ii. REPRESENTATIVE ELEMENTS:

The elements of sub group A is representative elements. They include metals, non-metals and metalloids. They are further classified into two blocks;

a) S-BLOCK ELEMENTS:

In which valence electron occupy ns^{1-2} orbital.

Or The elements in which the last electron enters in s orbital.

b) P-BLOCK ELEMENTS:

In which valence electron occupy np orbital.

Or The elements in which the last electron fills in np^{1-6} orbitals.

iii. TRANSITION ELEMENTS:

In periodic table elements of sub-group 'B' are known as transition or heavy elements. They are further divided into two categories:

a. d- BLOCK ELEMENTS (OUTER TRANSITION ELEMENTS):

The elements in which the last electron enters in $ns^2, (n-1) d^{1-10}$ orbital.

b. f- BLOCK ELEMENTS (INNER TRANSITION ELEMENTS):

The elements in which the last electron enters in $ns^2, (n-1) d^1, (n-2) f^{1-14}$ orbital.

VALENCE SHELL ELECTRONIC CONFIGURATIONS:

1. NOBLE GASES	ns^2, np^6
2. REPRESENTATIVE ELEMENTS	s-block elements: ns^{1-2} p-block elements: ns^2, np^{1-6}
3. TRANSITION ELEMENTS	d-block elements: $ns^2, (n-1)d^{1-10}$ f-block elements: $ns^2, (n-1)d^1, (n-2)f^{1-14}$

4. Define atomic radius? Describe the general trend in atomic size for groups and periods.

ATOMIC SIZE OR ATOMIC RADIUS

DEFINITION: The atomic radius is defined as one-half the distance between the nuclei of identical atoms that are bonded together.

OR

The total distance b/w the nucleus and the outer most orbit of the atom is called atomic radius.

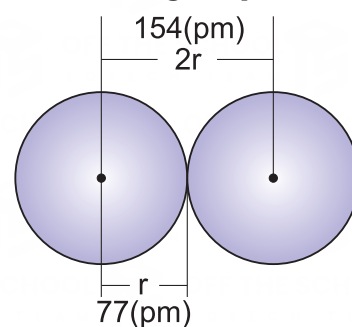
UNIT: it is measured in angstrom [$1\text{\AA} = 10^{-8}\text{ m}$]

PERIODIC TREND:

The atomic number increases from left to right across the period the atomic radius decrease. This gradual decrease in the radius is due to increase in the positive charge in the nucleus. As the positive nuclear charge^[1] increases, the negatively charged electrons in the shells are pulled closer to the nucleus. Thus, the size of the outermost shell becomes gradually smaller.

GROUP TREND:

In the periodic table, that atomic radius increases from top bottom within a group due to increase in atomic number. This is because of the addition of an extra shell of electrons in each element as we looking down in a group.



▲ The radius of carbon atom.

5. What is I.P? Describe the trends that exist in the periodic table for ionization energy.

IONIZATION POTENTIAL/ENTHALPY/ENERGY

DEFINITION: "The amount of energy which is required to remove an electron from an neutral, isolated, gaseous atom."

UNIT: It is measured in KJ/mol.

PERIODIC TREND:

Moving from left to right across the periodic table, the ionization energy for an atom increases. We can explain this by considering the nuclear charge of the atom. The more protons in the nucleus, the stronger the attraction of the nucleus to electrons. This stronger attraction makes it more difficult to remove electrons.

GROUP TREND:

Within a group, the ionization energy decreases as the size of the atom gets larger. We see that the ionization energy increases as we go up the group to smaller atoms. In this situation, the first electron removed is farther from the nucleus as the atomic number (number of protons) increases. Being farther away from the positive attraction makes it easier for that electron to be pulled off.

6. Explain shielding effect?

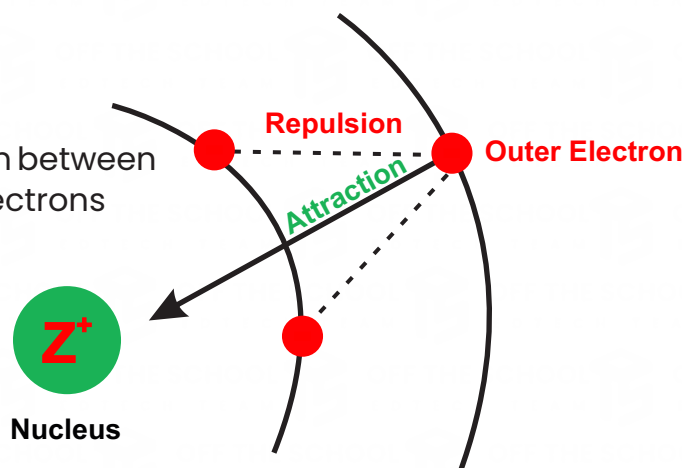
SHEILDING EFFECT

The shielding effect is the decrease in attraction between the nucleus and outer electrons due to inner electrons blocking the full nuclear charge.

7. Give briefing about electron affinity?

ELECTRON AFFINITY:

"The amount of energy released when an electron is added in the outermost shell of a neutral atom."



¹ The number of protons in the nucleus (called the nuclear charge).



UNIT: It is measured in KJ/mol or KJ.mol⁻¹ or KJ/mol.

PERIODIC TREND:

In a period, electron affinity increases from left to right due to decrease of atomic size because when size of atom increases the attraction between nucleus and incoming electron increases and more energy released.

GROUP TREND:

In a group electron affinity values decrease from Top to bottom, because the size of atom increases.

8. Do you have any idea about electronegativity? Describe the trends that exist in the periodic table for electronegativity?

ELECTRONEGATIVITY

DEFINITION: "The ability of an atom to attract the shared pair of electrons towards itself in a molecule is called electronegativity."

UNIT: Its S.I unit is Pauling-unit.

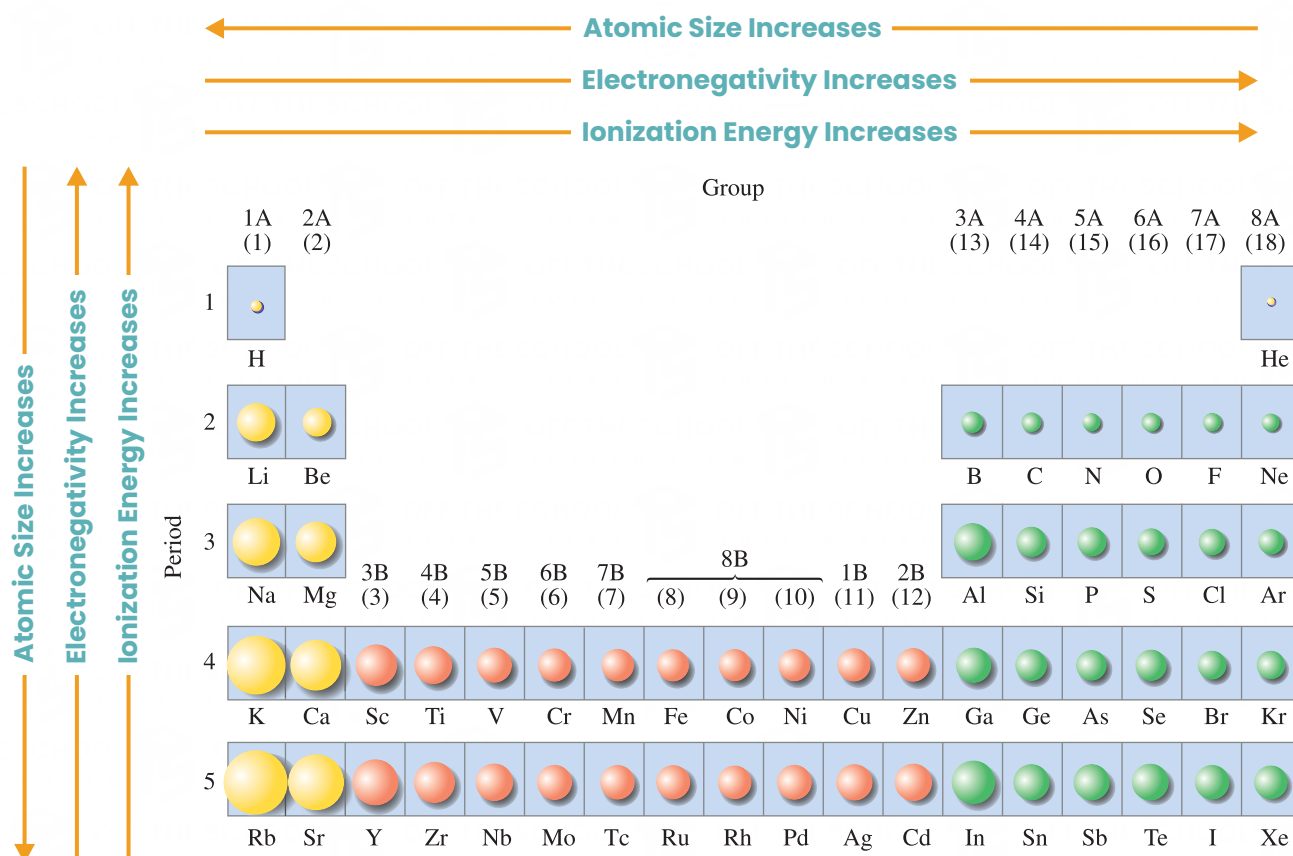
PERIODIC TREND:

Electronegativities generally increase from left to right across a period. This is due to an increase in nuclear charge. Alkali metals have the lowest electronegativities, while halogens have the highest. Because most noble gases do not form compounds, they do not have electronegativities.

GROUP TREND:

Electronegativities generally decrease from top to bottom within a group due to the larger atomic size.

The main group elements, fluorine has the highest electronegativity (EN = 4.0) and cesium the lowest (EN = 0.79).



PERIODS, GROUPS AND THEIR CHARACTERISTICS:

PERIOD	CHARACTERISTICS
1 ST	First This period has only two elements, hydrogen and helium which have only one energy level or shell. This period is called very-short period.
2 ND	Second It has 8 elements: Li 3 to Ne 10. Elements have two shells. The elements of this period are called bridge elements.
3 RD	This period also has 8 elements: Na 11 to Ar 18. The second and third periods are called short periods. These elements have 3 shells. The elements of this period are called typical elements.
4 TH	This period has 18 elements: K 19 to Kr 36. They have 4 energy levels or 4 shells.
5 TH	This period also has 18 elements: Rb 37 to Xe 54. They have 5 energy levels or 5 shells.
6 TH	This period has 32 elements: Cs 55 to Rn 86. They have six energy levels or 6 shells. The fourth, fifth and sixth periods are called long periods.
7 TH	Seventh This period has 32 elements. All these elements are radioactive. Out of these, the naturally occurring radioactive elements are Fr-87, Ra-88, Ac-89 Th-90, Po-91, and U-92 while the remaining elements after uranium, i.e., Np-93 to Db-105 are artificially prepared radioactive elements and these are called transuranic elements. Atoms of these elements have 7 energy levels or 7 shells.

Previous Year's Questions



1. An atom has only one electron in its M shell then what would be its atomic number? Identify the element and write four common characteristics of the group which it belongs to. **[2023]**
2. Identify the highly electronegative group of periodic table. Enlist the elements present in the Periodic table. Enlist the elements present in this group. Give three common characteristics of these elements. **[2024]**
3. How elements are arranged in periodic table? Describe groups & periods in the periodic table. **[2025]**

PERIODIC TABLE PACKET

Directions: Answer the questions with the proper information using your notes, book, and the periodic table.

1. Define a family. _____
2. What is a period? _____
3. What is the symbol for the following elements.
 - a. Magnesium _____
 - b. Potassium _____
 - c. Iron _____
 - d. Copper _____
4. What are the names of the following elements.
 - a. C _____
 - b. Cl _____
 - c. Au _____
 - d. Sr _____
5. What period are the following elements in?
 - a. He _____
 - b. Ge _____
 - c. Rb _____
 - d. I _____
6. What group are the following elements?
 - a. Sulfur _____
 - b. Ca _____
 - c. Iodine _____
 - d. Fe _____
7. Give me an atom with the following characteristics.
 - a. Halogen _____
 - b. Chalcogen _____
 - c. Alkali metal _____
 - d. Boron _____
 - e. Lanthanide series _____
 - f. Alkaline Earth metal _____
 - g. Transition metal _____
 - h. Noble gas _____

Directions: Use your Periodic table to complete the worksheet.

1. What is the atomic symbol for silver?
2. What is the atomic mass of mercury?
3. Ni is the symbol for what element?
4. The element that has the atomic number 17 is?
5. List the symbols for two transition metals.
6. Cu, Ag, and Au are all in what group #
7. Name two noble gases
8. Give the symbol for two halogens.
9. What is the symbol for element with atomic number 74?
10. What is the atomic mass of copper?
11. What is the last element in period 4?

For questions 12 – 15, label the following Key box as it should appear on your periodic table

12. _____
13. _____
14. _____
15. _____

6

C

Carbon

12.01

Periodic Table of Elements									
Directions: Use the periodic table to fill in the below chart.									
	Element	Symbol	Atomic Number	# of protons	# of electrons	Atomic Mass	Rounded Atomic Mass	(show work) # of Neutrons	Period
1	Oxygen	O	8	8	8	15.999	16	$16 - 8 = \underline{8}$	2
2	Helium								
3	Carbon								
4	Aluminum								
5	Calcium								
6	Sodium								
7	Potassium								
8	Nitrogen								
9	Silicon								
10	Iron								
11	Hydrogen								

Directions: Use a Periodic table to find the information asked for below:

1. What is the atomic number of:

Calcium _____
 Iron _____
 Gold _____
 Uranium _____

2. What is the Atomic mass of:

Calcium _____
 Iron _____
 Uranium _____
 Copper _____

3. How many protons do the following have?

Calcium _____

Gold _____

Copper _____

Iron _____

4. How many electrons do the following have?

Gold _____

Iron _____

Copper _____

Uranium _____

5. Does mercury have more protons and electrons than tin?

6. Is mercury a heavier element than tin?

7. Does potassium have more electrons than neon?

8. Does hydrogen have more electrons than Uranium?

9. Which has more protons, sulfur or iodine?

10. Write the symbols or the names for each of these elements:

Chlorine _____

Copper _____

Potassium _____

Silver _____

Na _____

Sn _____

Zn _____

Helium _____

Iron _____

P _____

Ne _____

Mercury _____

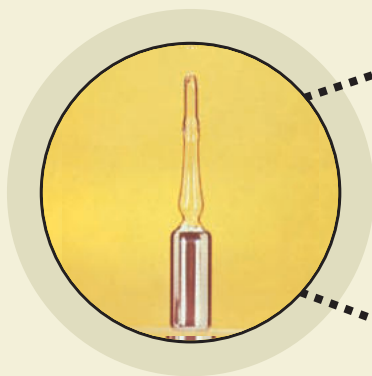
GROUP IA



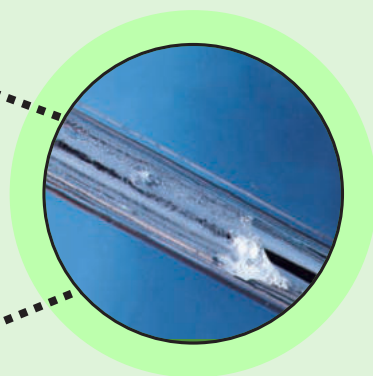
Lithium (Li)



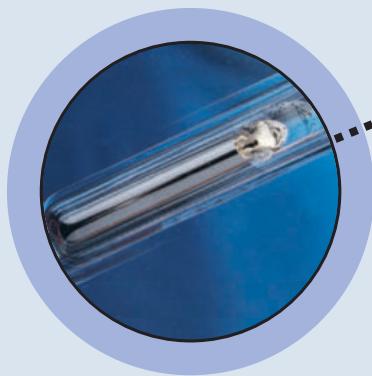
Sodium (Na)



Potassium (K)

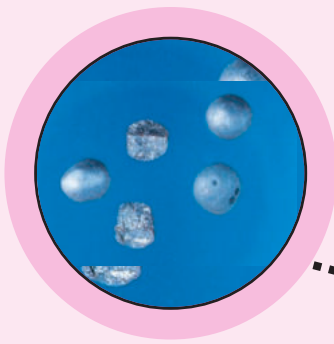


Rubidium (Rb)

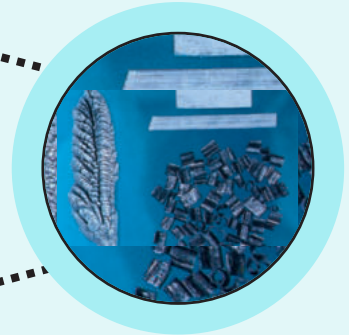


Cesium (Cs)

GROUP IIA



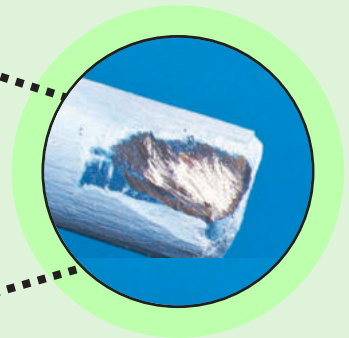
Beryllium (Be)



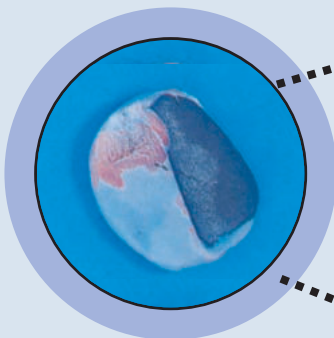
Magnesium (Mg)



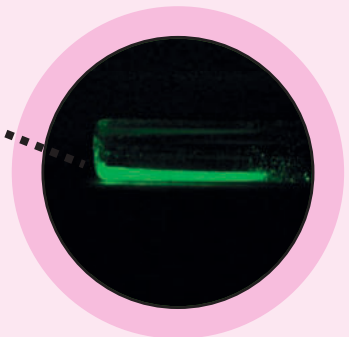
Calcium (Ca)



Strontium (Sr)

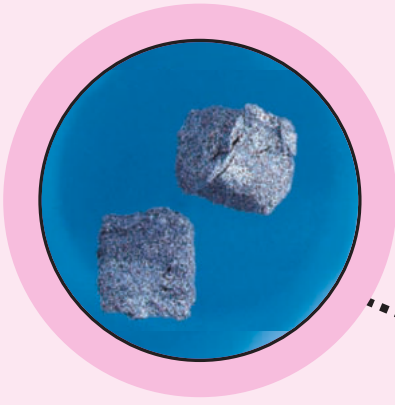


Barium (Ba)

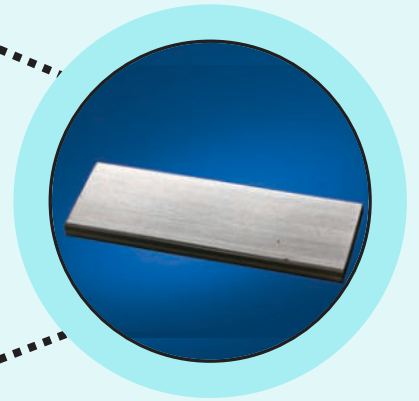


Radium (Ra)

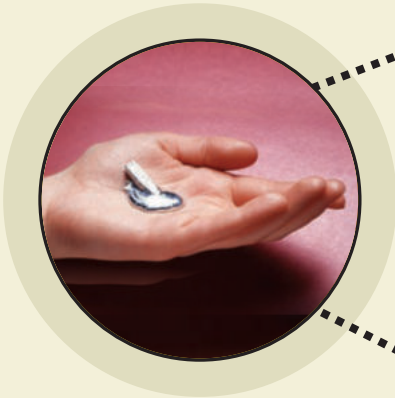
GROUP IIIA



Boron (B)



Aluminum (Al)

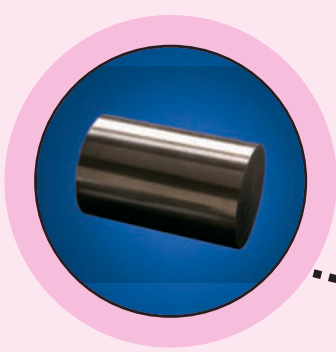


Gallium (Ga)



Indium (In)

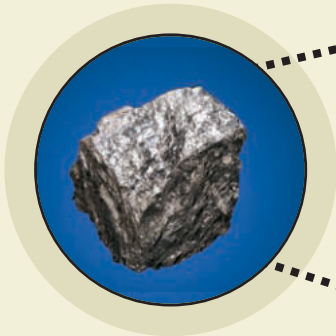
GROUP IVA



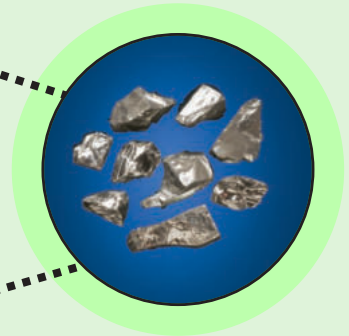
Carbon
(graphite)



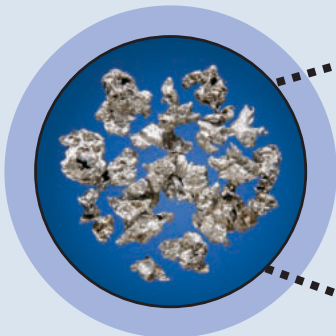
Carbon
(diamond)



Silicon (Si)



Germanium
(Ge)



Tin (Sn)



Lead (Pb)

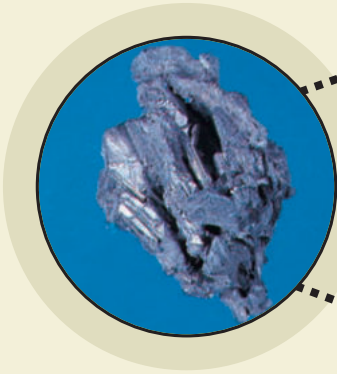
GROUP VA



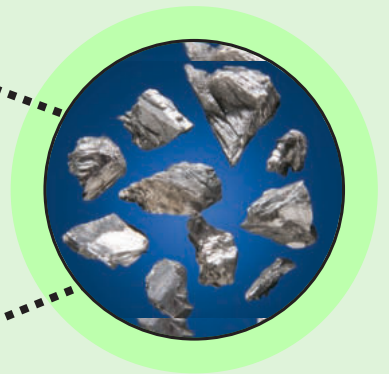
Liquid nitrogen (N_2)



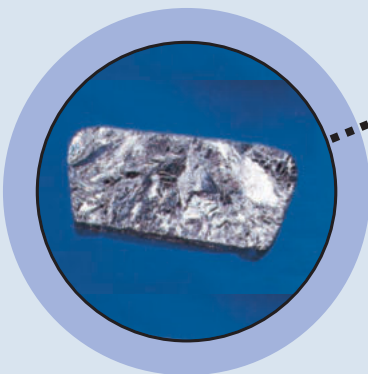
White and red phosphorus (P)



Arsenic (As)

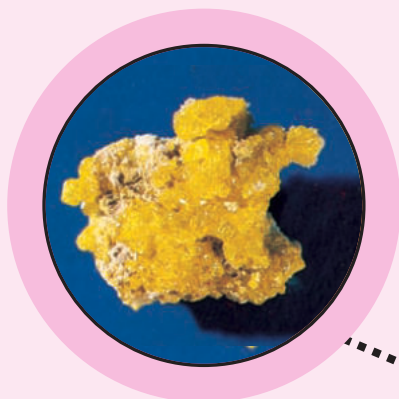


Antimony (Sb)

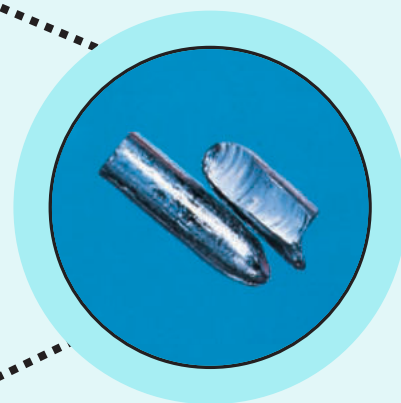


Bismuth (Bi)

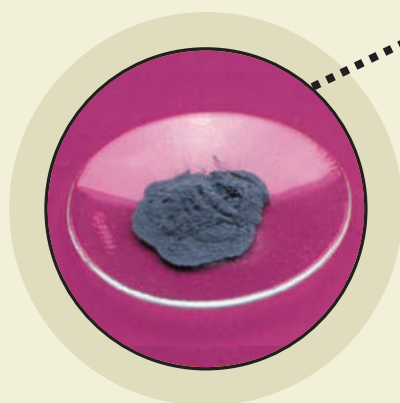
GROUP VIA



Sulfur (S₈)



Selenium (Se₈)



Tellurium (Te)

GROUP VIIA







Fluorine is a greenish-yellow gas that attacks ordinary glassware.
Astatine is radioactive.

GROUP VIIIA



Helium (He)



Neon (Ne)



Argon (Ar)



Krypton (Kr)



Xenon (Xe)

All noble gases are colorless and odorless.
These pictures show the colors emitted by the gases from a discharge tube.